

CS5788 Project Proposal

Mitigating Toxicity: A Comparative Study of Supervised Fine Tuning (SFT), Direct Preference Optimization (DPO), and Odds Ratio Preference Optimization (ORPO)

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1 Problem Statement

Large Language Models (LLMs) often generate toxic or harmful content due to unfiltered pre-training data. Aligning these models using Reinforcement Learning from Human Feedback (RLHF) is essential for safety. However, traditional methods are computationally expensive. This project focuses on comparing two offline preference optimization methods: **Direct Preference Optimization (DPO)**, which requires a reference model, and **Odds Ratio Preference Optimization (ORPO)**, a reference-free alternative.

The core challenge is evaluating these methods under resource-constrained conditions. Since DPO requires maintaining two models in memory, it presents a significant hardware bottleneck compared to ORPO. Beyond toxicity reduction, our team will also analyze the trade-offs between computational overhead and model performance, specifically whether ORPO can reduce toxicity as effectively as DPO while maintaining the high fluency and quality.

2 Related Work

- **RLHF for Alignment:** Reinforcement Learning from Human Feedback is widely used to align large language models with human preferences for safety and helpfulness. Bai et al. [3] propose the Helpful-Harmless (HH-RLHF) framework, which uses supervised fine-tuning, reward modeling, and reinforcement learning. While effective, this pipeline is computationally expensive and difficult to scale.
- **DPO:** Rafailov et al. [1] reformulate preference learning as a direct optimization problem, eliminating the need for a reward model. By using a frozen reference model for regularization, DPO achieves stable alignment without reinforcement learning, but requires additional memory and computation to maintain the reference model.
- **ORPO:** Hong et al. [2] propose a reference-free alternative that integrates preference learning with language modeling. ORPO combines supervised fine-tuning with an odds-ratio-based objective, improving efficiency. However, its effectiveness in safety-critical tasks such as toxicity reduction remains underexplored.

Our Contribution. We directly compare DPO and ORPO under a unified framework, focusing on their effectiveness in toxicity mitigation and their computational trade-offs in resource-constrained settings.

3 Methodology

3.1 Dataset:

- **Training:** Initially, we decide to utilize the **Anthropic HH-RLHF** dataset [3]. This dataset consists of two primary fields: **chosen** and **rejected**, and each of them represents a full dialogue transcript between a **Human** and an **Assistant**. The chosen field contains the response considered as safer and more helpful, while the rejected field contains the toxic responses. We will split the raw dialogue into three parts: the **prompt**, the **chosen_response**, and the **rejected_response**. This "triplet" format lets the model learn by comparing the two answers directly.

3.2 Architecture / Algorithms:

- **Base Model Selection:** Due to DPO’s requirement of maintaining both a trainable policy and a frozen reference model in memory, we must select a small-scale, decoder-only Transformer. Potential candidates include **GPT-2 Small (124M)** as a standard baseline, **GPT-2 Medium (355M)** for improved generation quality, and **Pythia-410M** for its modern, reproducible training framework.
- **Algorithm Implementation:**
 - **Direct Preference Optimization (DPO) [1]:** We will implement DPO by optimizing the log-likelihood ratio between the current policy π_θ and a frozen reference model π_{ref} . This directly increases the probability of the chosen response and penalizes the rejected one, normalized by the base distribution.
 - **Odds Ratio Preference Optimization (ORPO) [2]:** ORPO is a memory-efficient alternative that eliminates the reference model requirement. It achieves preference alignment using a combined objective, $Loss_{ORPO} = Loss_{SFT} + \lambda \cdot Loss_{OR}$, which uses an odds ratio penalty ($Loss_{OR}$) to penalize rejected responses while maintaining generative fluency ($Loss_{SFT}$).

3.3 Evaluation:

We will perform a comparison between the baseline SFT model, the DPO-aligned model, and the ORPO-aligned model. Using the **RealToxicityPrompts** dataset [4], which consists of the following fields: **filename**, **begin**, **end**, **challenging**, **prompt**, and **continuation**. Specifically, we will extract the input prefixes from the **text** key within the **prompt** field, prioritizing samples where **challenging** is **True** to elicit toxic responses. The models’ generated continuations will then be evaluated by a pre-trained Toxic-BERT classifier (e.g., **unitary/toxic-bert**), which outputs a probability score ($s \in [0, 1]$) used to calculate both the Average Toxicity Score and the Probability of Toxicity.

4 Deliverables and Expected Results

During the training and evaluation processes, we will compare the results of the baseline SFT model, the DPO-aligned model, and the ORPO model. The SFT model is expected to outperform the pre-trained model, while the DPO-aligned and ORPO models are anticipated to demonstrate greater robustness when handling challenging samples. In particular, these models should be less likely to produce toxic or inappropriate responses after being trained with our proposed approach.

5 Feasibility

We plan to employ relatively small models, with no more than 7–13B parameters, to ensure that our methodology remains feasible in resource-constrained scenarios and to significantly reduce GPU memory for both training and testing. Based on scaling law, we recognize that model performance improves with increases in parameter size and data volume. If the current model size proves insufficient, we will consider transitioning to larger architectures and expanding our data collection accordingly. To support these more computationally intensive requirements, we plan to leverage Google Colab’s high-performance hardware resources to ensure adequate computational capacity.

6 Workload Distribution

- **Leyan Wang:** Select the project topic and then evaluate the feasibility of training, inference, and testing procedures of models.
- **Zijing Wu:** Implement the data preprocessing pipeline for the HH-RLHF dataset and implement the final toxicity evaluation scripts using Toxic-BERT.
- **Yan Xiao:** Implement the training code to establish the SFT baseline and conduct the actual fine-tuning processes for both DPO and ORPO models.

References

- [1] R. Rafailov, A. Sharma, E. Mitchell, S. Ermon, C. D. Manning, and C. Finn, “Direct Preference Optimization: Your Language Model is Secretly a Reward Model,” in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2023.
- [2] J. Hong, N. Lee, and J. Thorne, “ORPO: Monolithic Preference Optimization without Reference Model,” *arXiv preprint arXiv:2403.07691*, 2024.
- [3] Y. Bai et al., “Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback,” *arXiv preprint arXiv:2204.05862*, 2022. [Online]. Available: <https://huggingface.co/datasets/Anthropic/hh-rlhf>
- [4] S. Gehman, S. Gururangan, M. Sap, Y. Choi, and N. A. Smith, “RealToxicityPrompts: Evaluating Neural Toxic Degeneration in Language Models,” in *Proc. Find. Assoc. Comput. Linguist. (EMNLP)*, 2020. [Online]. Available: <https://huggingface.co/datasets/allenai/real-toxicity-prompts>
- [5] T. Davidson, D. Warmusley, M. Macy, and I. Weber, “Automated Hate Speech Detection and the Problem of Offensive Language,” in *Proc. ICWSM*, 2017.
- [6] Unitary AI, “Detoxify: Toxic Comment Classification,” <https://github.com/unitaryai/detoxify>